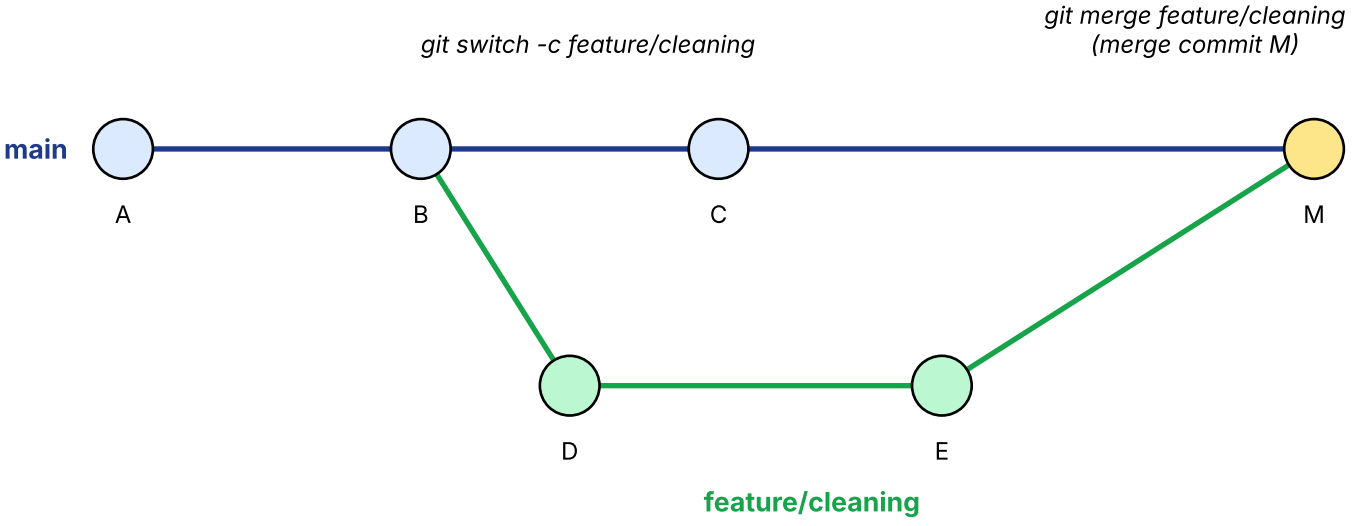Branching and merging